# Length of a Line

Exam code: 0607

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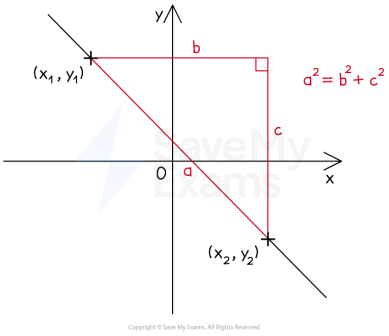
## Length of a line

### How do I calculate the length of a line?

- The **distance between two points** with coordinates $(x_1, y_1)$ and $(x_2, y_2)$ can be found using the formula

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- This formula uses **Pythagoras' theorem** $a^2 + b^2 = c^2$
  - It is applied to the difference in the $x$-coordinates and the difference in the $y$-coordinates




Help me understand

### Examiner Tips and Tricks

Be extra careful when negative coordinates are involved. It can help to put negative numbers in brackets to make your working clearer, e.g. $(-6) - (-8) = 2$.

### Worked Example

Point *A* has coordinates (3, -4) and point *B* has coordinates (-5, 2).

Calculate the distance of the line segment *AB*.

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Ask questions and get help with exactly what you're revising.
Got it

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Substitute in the two given coordinates

$$d = \sqrt{(3-(-5))^2 + ((-4)-2)^2}$$

Be careful with the negative numbers
$3-(-5) = 8$ and $(-4)-2 = -6$

Simplify

$$d = \sqrt{(8)^2 + (-6)^2} = \sqrt{64+36} = \sqrt{100} = 10$$

**10 units**

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